



Q1: [50] Consider a quarter-circle slab. It is composed of two materials with permittivities ϵ_1 and ϵ_2 . The two faces of the slab, $\rho = a$ and $\rho = c$, are grounded; they have zero electric potential. In the middle at $\rho = b$ some free charges are implanted with surface charge density ρ_S . Solving Laplace's equation, find the values of electric potential V and electric field intensity $\mathbf{E}$ in materials 1 and 2.

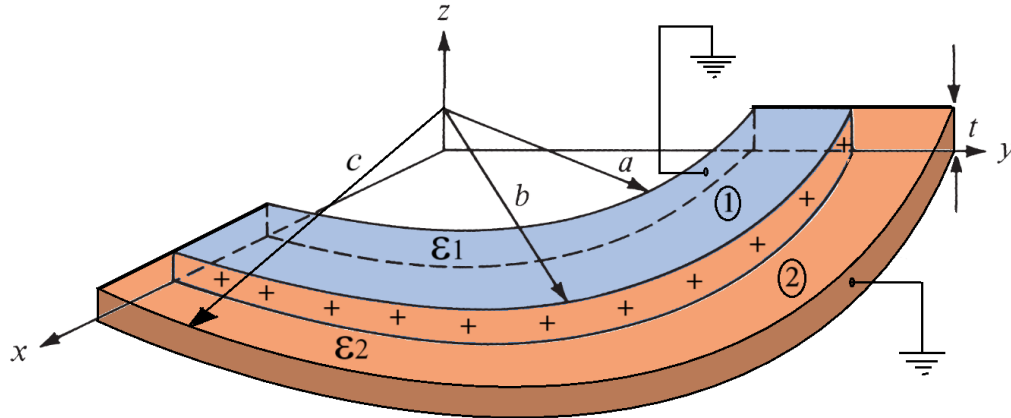


Figure 1: Fig. for Question 1.

Q2: [50] Using the method of images, find the value of the electric field and electric potential at the field-point at $(0, 0, h/2)$ over the perfect conducting plane in the presence of a point charge Q at $(0, d, h/2)$, and a line charge parallel to the y -axis passing through the z -axis at $z = h$;

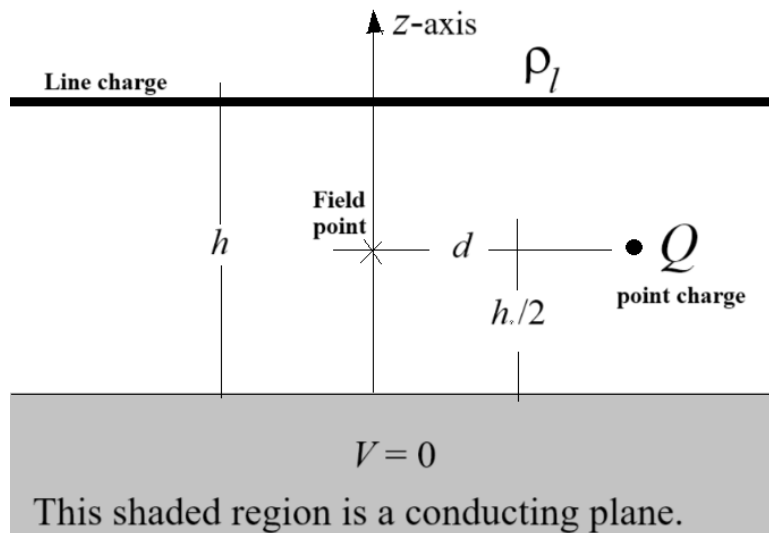


Figure 2: Fig. for Question 2.